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1 Aim of the Experiment

The primary objective of this experiment was to recover an On-Off Keying (OOK) signal at the receiver and quantitatively measure how additive noise affects the bit error probability (P_b).

The specific deliverables and methodology included:

- Rebuilding and tuning the analog OOK transmitter.
- Characterizing the receiver band-pass filter's amplitude response.
- Observing the time-domain signals across the demodulator, matched filter (integrator), and sample-and-hold (S&H) stages.
- Calibrating the analog offset and decision threshold for optimal symbol detection.
- Injecting controlled noise to plot the experimental Bit Error Ratio (BER) against the Signal-to-Noise Ratio (SNR) and comparing it to theoretical models.

2 Theoretical Background

In OOK, the digital baseband signal modulates the amplitude of a continuous carrier wave; the carrier is fully transmitted to represent a digital '1' and completely suppressed to represent a digital '0'. At the receiver, coherent demodulation down-mixes the signal.

To maximize the SNR at the sampling instant, the system utilizes a matched filter, implemented here as a short-term integrator. The integrator accumulates the signal energy over exactly one symbol period, effectively averaging out zero-mean random noise. At the end of each symbol period T_s , the final integrated voltage is sampled and held constant while the integrator capacitor is rapidly discharged to prepare for the next symbol.

A decision circuit (comparator) evaluates this held voltage against a defined threshold. For ideal S&H levels of 0 V and 1.0 V, the optimal decision threshold lies exactly at the midpoint (0.5 V). The bit error probability is defined statistically as:

$$P_b = \frac{\text{Number of incorrectly received bits}}{\text{Total number of received bits}}$$

3 Equipment and Experimental Setup

The physical channel and detection system utilized the following hardware components:

- DAM Modem Board
- Pseudo-Random Number Generator (PZG, 4 kHz clock)
- Function Generator
- 4-Channel Oscilloscope
- Audio Analyzer (UPV)
- Noise Generator
- RMS Voltmeter
- External Adder and External 20 dB Attenuator
- Bit Error Ratio Meter (BERM)

The completely assembled signal path was structured as follows:

PZG → OOK Modulator → 20 dB Attenuator → Adder
 → Band-Pass Filter → Demodulator → Integrator (Matched Filter)
 → Sample-and-Hold → Decision Circuit (Comparator)



Figure 1: DAM board connections highlighting the clock input, sampling-point control, and decision-level control adjustments.

4 Measurements and Results

4.1 Task 4.1: OOK Transmitter Setup

Setup

The analog transmission side from the previous lab was rebuilt. The function generator was configured to supply a continuous sine wave carrier to the modulator. The **Mod-In-Offset** potentiometer on the DAM board was manually tuned while monitoring the **RF_OUT** socket on the oscilloscope to achieve OOK modulation.

Desired Values (Theory)

For ideal OOK modulation, the RF output should display a full-amplitude sinusoidal carrier wave when the digital input (**Mod_IN**) is logic HIGH. When the input is logic LOW, the carrier should be completely suppressed (0 V flatline).

Collected Data

The function generator was initially set to $f_c = 20 \text{ kHz}$ and $U_{pp} = 0.3 \text{ V}$. After confirming the filter characteristics in the subsequent task, both transmitter and receiver carrier frequencies were updated to align with the band-pass filter's center frequency: $f_c = 58.125 \text{ kHz}$.

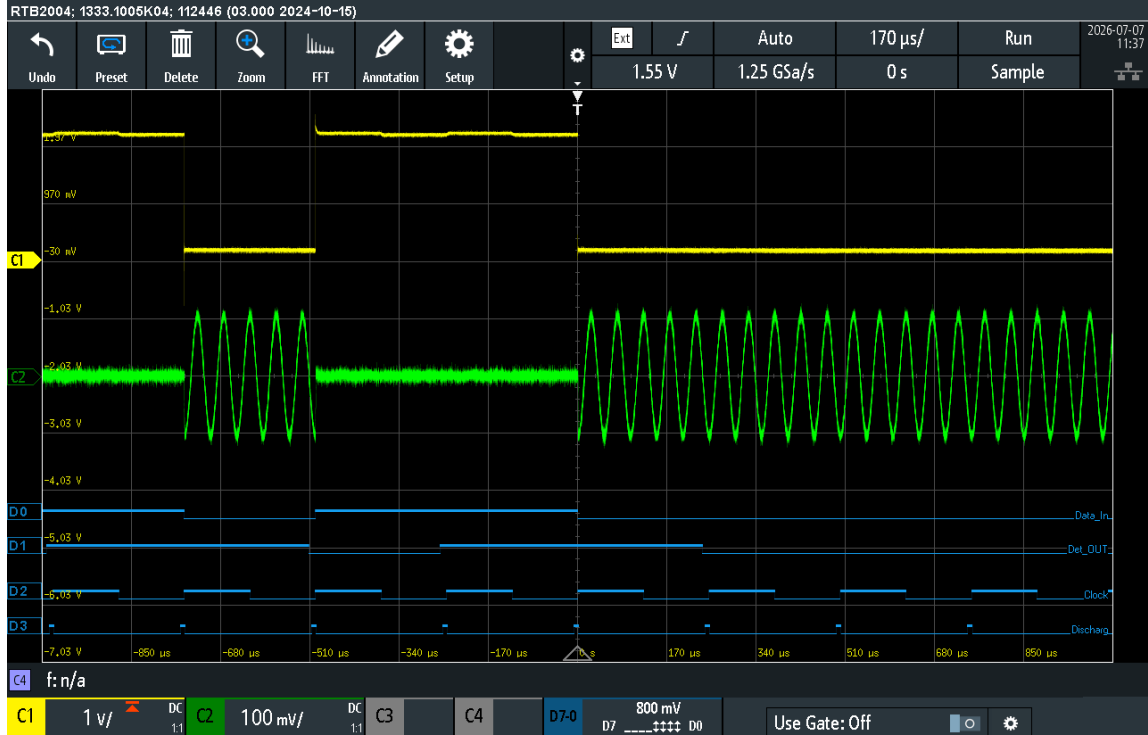


Figure 2: OOK signal at the modulator output. The carrier is present only during the active symbols (logic HIGH), confirming successful offset calibration.

Analysis & 'The Why'

The **Mod-In-Offset** adjustment is strictly required because analog multipliers process voltages literally. By shifting the DC offset of the digital baseband signal, we ensure that a digital '0' equates to precisely 0 V entering the multiplier, which results in zero carrier output ($0 \times \text{Carrier} = 0$).

4.2 Task 4.2: Band-Pass Filter Measurement

Setup

The amplitude response of the receiver's band-pass filter was measured using the UPV Audio Analyzer. A reference measurement was taken by connecting the UPV output directly to its input. The filter was then inserted into the measurement loop (UPV OUT $\rightarrow$ BP_IN $\rightarrow$ BP_OUT $\rightarrow$ UPV IN).

Desired Values (Theory)

The filter should exhibit a center frequency (f_M) closely matching the updated carrier frequency, with a 3 dB bandwidth wide enough to pass the 16 kHz useful signal band (approx. $f_M \pm 8$ kHz) without significant amplitude distortion.

Collected Data

The maximum response was located at $f_M = 58.125$ kHz with $A_{\max} = -5.018$ dB. The measured 3 dB cut-off frequencies ($A = -8.018$ dB) were:

$$f_L = 33.948 \text{ kHz}, \quad f_H = 99.518 \text{ kHz}$$

Resulting in a 3 dB bandwidth of:

$$B_{3\text{dB}} = f_H - f_L = 65.570 \text{ kHz}$$

Checking the required 16 kHz band ($f_M \pm 8$ kHz):

$$\begin{aligned} f_{\text{low}} &= 50.125 \text{ kHz}, & A(50.125 \text{ kHz}) &= -5.303 \text{ dB} \\ f_{\text{high}} &= 66.125 \text{ kHz}, & A(66.125 \text{ kHz}) &= -5.254 \text{ dB} \end{aligned}$$

Compared with the center value, the additional attenuation at the edges of the useful band was only $\Delta A_{\text{low}} = 0.285$ dB and $\Delta A_{\text{high}} = 0.236$ dB.

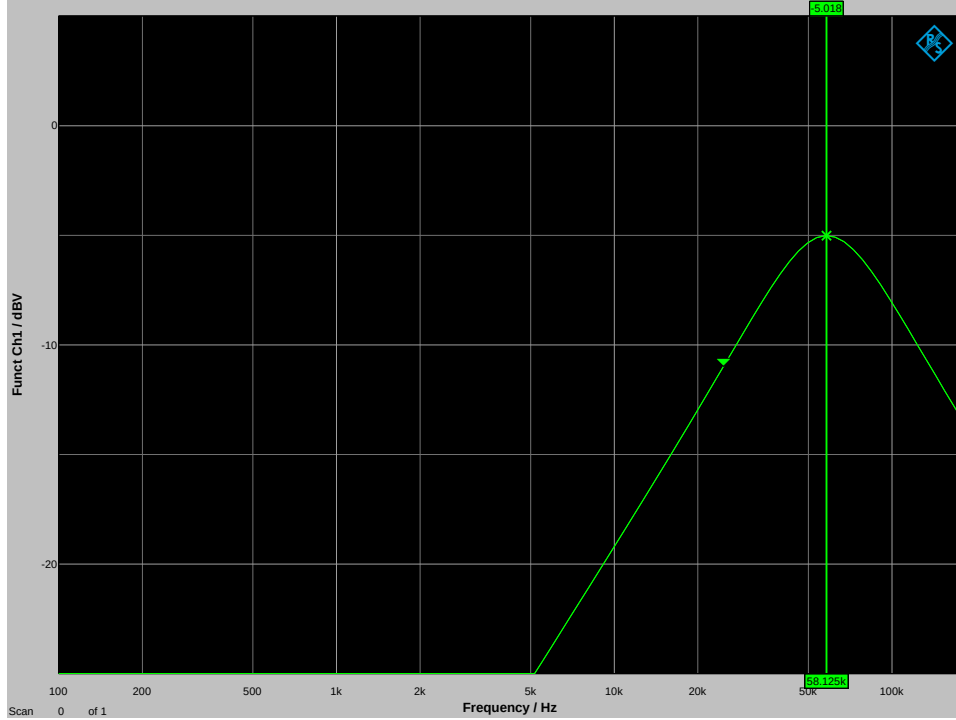


Figure 3: Measured band-pass response with the marker identifying the center frequency at 58.125 kHz and A_{max} of -5.018 dB.

Analysis & 'The Why'

The analysis confirms that the 16 kHz useful band passes almost entirely unattenuated (variance < 0.3 dB). This flat passband is critical for preserving signal integrity; if the sidebands were unevenly attenuated, the time-domain envelope of the OOK signal would distort, leading to intersymbol interference (ISI) and a degraded BER.

4.3 Task 4.3: Demodulator and Integrator Without Band-Pass Filter

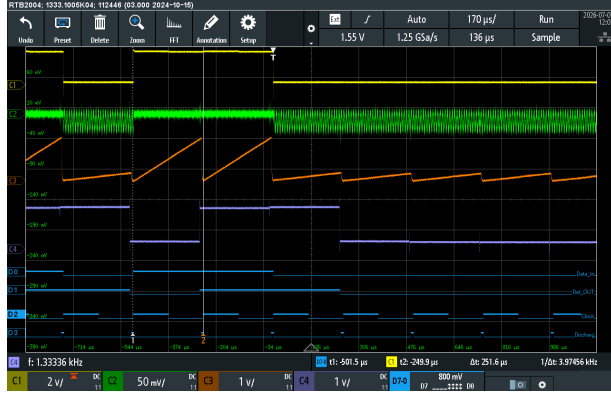
Setup

To verify the matched filter operation, the adder output was temporarily connected directly to the AUX input, bypassing the band-pass filter. Signals were mapped to the oscilloscope: CH1 (Mod_IN), CH2 (Demod_OUT), CH3 (Int_OUT), and CH4 (S&H_OUT).

Desired Values (Theory)

The integrator (matched filter) should accumulate voltage linearly during a logic HIGH symbol, creating a distinct ramp. At the end of the symbol period T_s , the capacitor must rapidly discharge to 0 V, while the S&H block captures and sustains the peak integrated value for the duration of the next symbol.

Collected Data



(a) Observation over approximately two symbols.



(b) Observation over approximately twenty symbols.

Figure 4: Receiver signals bypassing the band-pass filter. CH3 (orange) shows the integration ramps, and CH4 (purple) shows the held values.

Analysis

The matched filter behavior is clearly visible. The integrator linearly sums the energy of the demodulated signal. The integrator accumulates the demodulated signal over one symbol interval. For the approximately rectangular symbol waveform, this operation acts as a matched filter and improves the SNR at the sampling instant.

4.4 Task 4.4: Receiver Signals With Band-Pass Filter

Setup

The signal path was corrected to include the band-pass filter: **Adder Output** → **BP_IN** → **BP_OUT** → **AUX**.

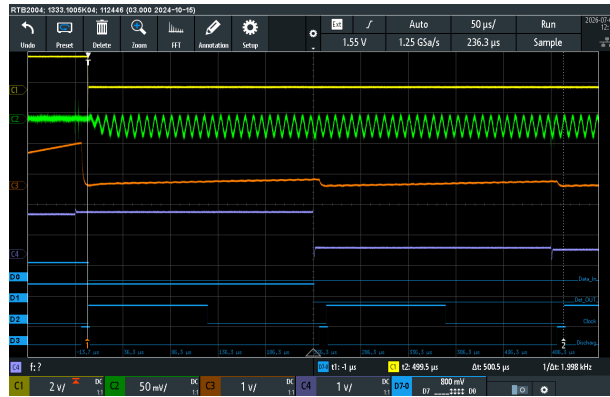
Desired Values (Theory)

Introducing the band-pass filter should smooth the sharp edges of the incoming square-wave modulated envelope and introduce a slight time delay, while successfully passing the fundamental carrier frequency.

Collected Data



Figure 5: CH1: Input signal before the band-pass filter. CH2: Output signal after the band-pass filter, showing envelope smoothing.



(a) Observation over approximately two symbols.



(b) Observation over approximately twenty symbols.

Figure 6: Receiver signals after inserting the band-pass filter.

Analysis & 'The Why'

The filter induces a visible delay and transient rounding (ringing) at the symbol transitions. This occurs due to the filter's inherent group delay and because rejecting out-of-band frequencies strips away the high-frequency harmonics required to make perfectly sharp square-wave edges. Despite these transients, the integrator still resolves two clearly separated sample-and-hold levels, proving the system is robust against this filtering distortion.

4.5 Task 4.5: Offset and Decision-Level Adjustment

Setup

The `Offset_Komp` potentiometer was tuned to center the integrator's base level. The receiver-carrier amplitude was adjusted to define the S&H levels. Finally, CH3 was moved to monitor the `Decision-Level` voltage, which was adjusted via its respective potentiometer.

Desired Values (Theory)

The integrator must not drift (indicative of DC offset). S&H output voltages should peak near 1.0 V for a '1' and 0 V for a '0'. The decision threshold must be set exactly equidistant from both levels (at 0.5 V) to minimize the probability of noise triggering a false decision in either direction.

Collected Data

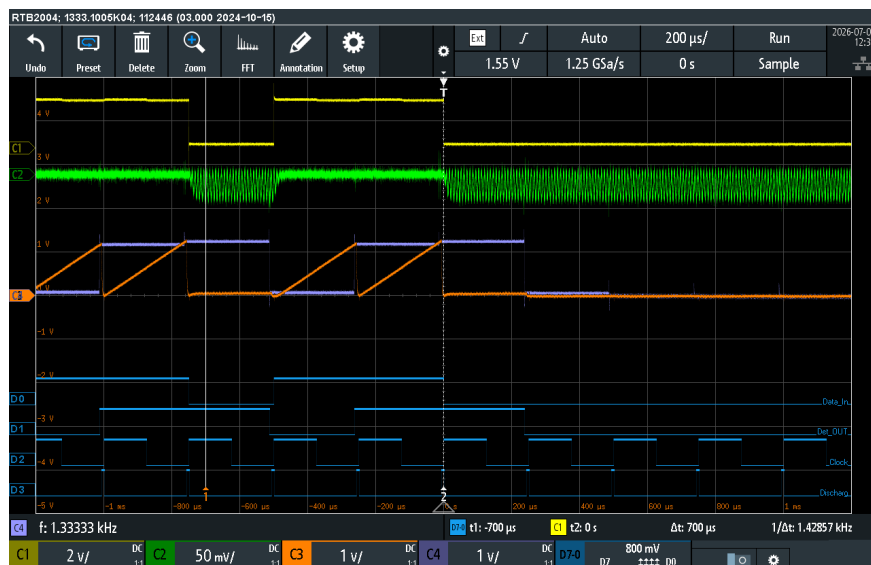


Figure 7: Receiver signals after offset compensation. The integrator returns to a stable 0 V baseline after every discharge, eliminating drift.

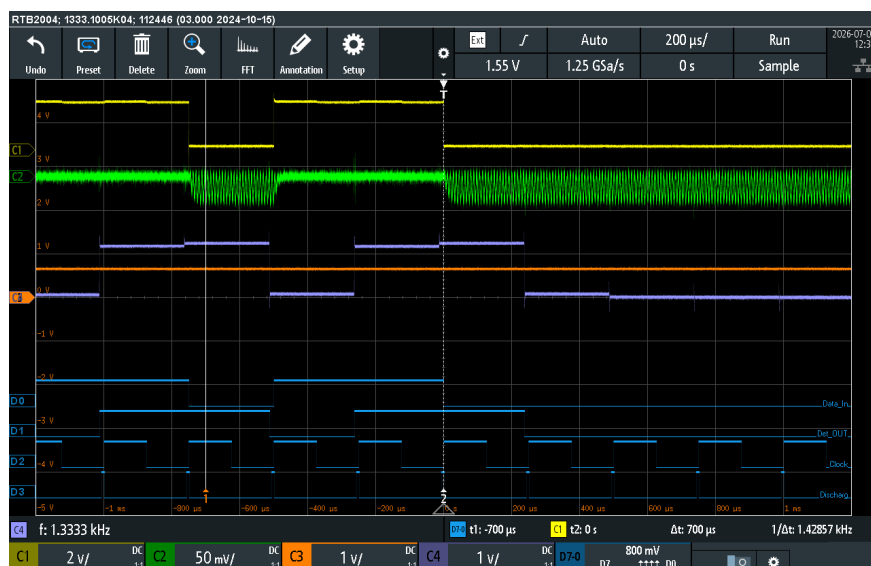


Figure 8: Decision-level adjustment for OOK. CH3 shows the constant `Decision-Level` threshold adjusted to 0.5 V, perfectly bisecting the S&H OOK levels.

Analysis & 'The Why'

Uncompensated DC offsets in analog mixers or op-amps will be integrated over time, causing the output to "ramp" uncontrollably (drift), which would force the S&H voltage out of the comparator's range. By zeroing the drift and setting the threshold exactly at 0.5 V, the system ensures maximum mathematical noise immunity. The resulting `Det_OUT` signal successfully mirrors the transmitted data, shifted only by the physical processing delay of the hardware chain.

4.6 Task 4.6: BER Calibration and Noise Injection

Setup

The SUF2 Noise Generator (110 kHz bandwidth) was connected to the free adder input. The BER meter was synchronized with the PZG clock, and the delay shift was adjusted to align the `Mod_IN` and `Det_OUT` streams. The RMS voltmeter measured effective voltages at the S&H output.

Desired Values (Theory)

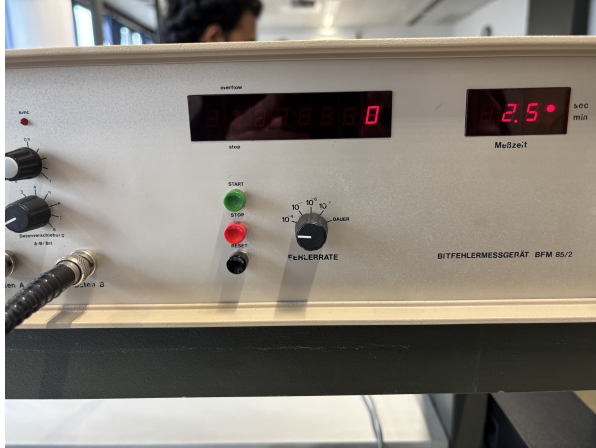
To map BER against SNR, a reference point must be established. The desired starting point is a high noise state yielding a Bit Error Ratio of $P_b \approx 0.25$. Crucially, the `AUX` signal amplitude must not exceed $2V_{pp}$ during this heavy noise injection.

Collected Data

The initial noiseless reference voltage was measured as $U_{E,0} = 0.88\text{ V}$. Noise was added (-22 dB setting) until the BER display (on the 10^{-4} range) read 2457, corresponding to:

$$P_b \approx 2457 \cdot 10^{-4} = 0.2457$$

The total RMS voltage with noise was measured as $U_{E,\text{ref}} = 1.25\text{ V}$. The `AUX` signal was monitored and confirmed to remain below the $2V_{pp}$ limit.



(a) BER meter after delay synchronization, showing zero errors without added noise.



(b) Noisy measurement on the 10^{-4} range, establishing the reference $P_b \approx 0.25$.

Figure 9: BER-meter calibrations before and after adding additive white noise.

Analysis

Verifying that the `AUX` signal remains below $2V_{pp}$ is a critical measurement validity protocol. The `AUX` input feeds the analog multiplier (mixer). If the combined signal-plus-noise exceeds this threshold, the multiplier is overdriven and enters non-linear clipping. Clipping destroys the linear mathematical relationship between signal power and noise power, rendering all subsequent SNR

derivations entirely invalid. By staying below this threshold, the integrity of the SNR calculation is preserved.

4.7 Task 4.7: BER Curve Measurement

Setup

The measurement started at the reference noise-generator setting of -22 dB. The noise level was then reduced in steps of 2 dB and later 1 dB. For the first measurements, the BER meter resolution was set to 10^{-4} . At lower error probabilities, the resolution was changed to 10^{-5} to obtain more reliable error counts.

Theory

Reducing the noise power by 2 dB increases the SNR by approximately 2 dB. Therefore, the bit error probability should decrease exponentially as the noise level is reduced.

Because SNR in this specific lab is defined via physical RMS power measurements (P_S/P_N) rather than theoretical energy per bit (E_b/N_0), the theoretical Q-function curves adjust accordingly. The theoretical BER of coherent OOK is:

$$P_b = Q\left(\sqrt{\frac{\text{SNR}}{2}}\right)$$

The theoretical BPSK curve is:

$$P_b = Q\left(\sqrt{\text{SNR}}\right)$$

Collected Data

The noiseless RMS voltage was $U_{E,0} = 0.88$ V, and the RMS voltage at the reference noise level was $U_{E,\text{ref}} = 1.25$ V.

The reference SNR was calculated mathematically:

$$\text{SNR}_{\text{ref}} = \frac{U_{E,0}^2}{U_{E,\text{ref}}^2 - U_{E,0}^2} = 0.9826$$

In decibels:

$$\text{SNR}_{\text{ref,dB}} = 10 \log_{10}(0.9826) = -0.076 \text{ dB}$$

Table 1: Measured BER values for different noise-generator settings.

Noise level (dB)	RMS voltage (V)	Range	Display	P_b
-22	1.25	10^{-4}	2457	0.2457
-24	1.10	10^{-4}	1960	0.1960
-26	1.00	10^{-4}	1424	0.1424
-28	0.98	10^{-4}	841	0.0841
-30	0.93	10^{-4}	447	0.0447
-32	0.91	10^{-4}	207	0.0207
-34	0.88	10^{-4}	31	0.0031
-36	0.88	10^{-4}	2–6	$(2\text{--}6) \times 10^{-4}$
-34	0.90	10^{-5}	412	0.00412
-35	0.89	10^{-5}	140–170	$(1.40\text{--}1.70) \times 10^{-3}$
-36	0.88	10^{-5}	36	0.00036

The measurements at -34 dB and -36 dB were repeated using the finer 10^{-5} range. These values were used for the final BER plot. For the -35 dB measurement, the average display value of 155 was used.

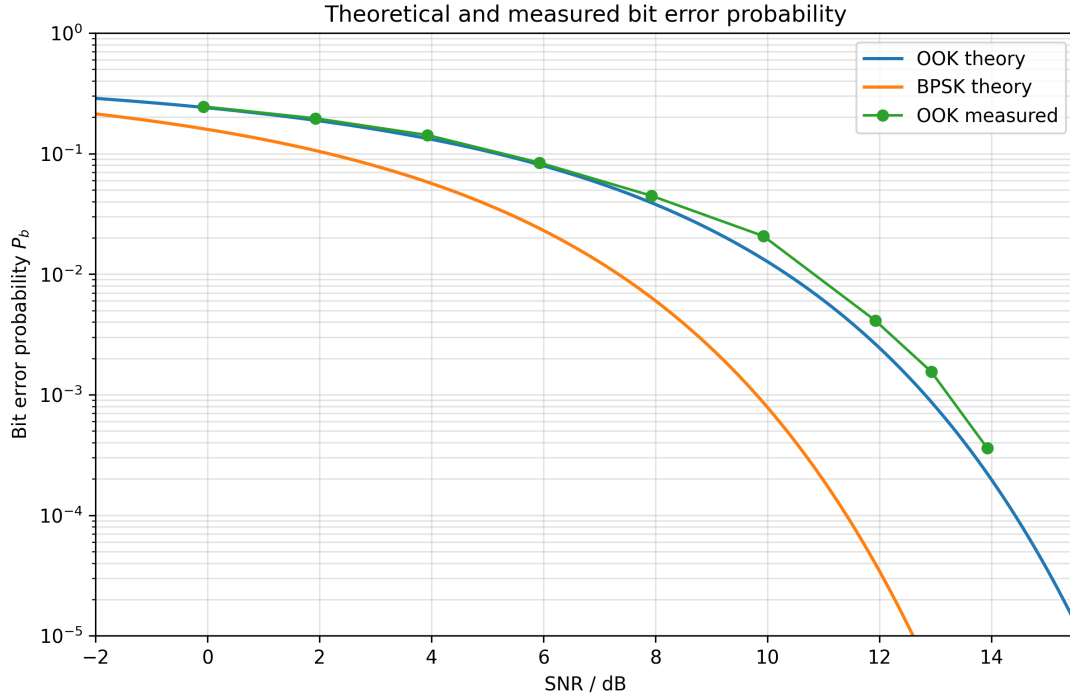


Figure 10: Measured OOK BER compared with the theoretical OOK and BPSK curves.

Analysis

The measured bit error probability decreased continuously when the noise level was reduced. This agrees exactly with the expected behaviour because a lower noise power results in a higher SNR.

At -36 dB, only 2 to 6 errors were observed on the 10^{-4} range. This number was too small for a statistically reliable measurement. After changing the resolution to 10^{-5} , 36 errors were measured over the longer sample time, corresponding to $P_b = 3.6 \times 10^{-4}$.

The measured BER curve follows the theoretical OOK curve closely. Small differences (approx. 0.5 dB gap) are physically expected due to imperfect adjustment of the decision level, filter inter-symbol interference, phase synchronization differences, and innate analog component noise.

5 Discussion

The OOK transmitter and receiver were successfully adjusted and tested. The transmitter offset was set so that the carrier was present for a logic “1” and suppressed for a logic “0”.

The band-pass filter had its maximum response at $f_M = 58.125$ kHz. Across the required 16 kHz signal band, the amplitude changed by less than 0.3 dB relative to the center frequency. Therefore, the useful OOK sidebands passed through the filter with only minor frequency-dependent attenuation.

The integrator accumulated the demodulated signal over one symbol period. The sample-and-hold circuit then stored the final value, allowing the decision circuit to distinguish between the two symbol levels. The decision threshold was adjusted to approximately $U_D = 0.5$ V, which lies midway between the measured levels for logic “0” and logic “1”. This provided optimal noise immunity and allowed the detected output to faithfully reconstruct the transmitted data.

6 Conclusion

The analog OOK transmission system was successfully assembled, adjusted and measured. The carrier frequency was matched to the center frequency of the receiver band-pass filter, and the offset, sampling point and decision level were adjusted correctly.

Noise was added to the received signal, and the resulting bit error probability was measured across a range of noise levels. The BER decreased from $P_b = 0.2457$ at the highest noise level to $P_b = 3.6 \times 10^{-4}$ at the lowest measured noise level.

The measured results can be fully explained to the expected theoretical behaviour of coherent OOK. The remaining minor differences can be fully explained by the non-ideal properties of physical electronics, such as filter group-delay, imperfect thresholding, minor phase discrepancies, and the inherent analog noise floor.

A MATLAB Code Used for the BER Plot

```

1 % Calculation and Plotting of Experimental vs Theoretical BER
2 clc; clear; close all;
3
4 % Calibration Measurements (RMS Voltages)
5 Ue_0 = 0.88;
6 Ue_ref = 1.25;
7
8 % Calculate Reference SNR
9 SNR_ref = Ue_0^2 / (Ue_ref^2 - Ue_0^2);
10 SNR_ref_dB = 10 * log10(SNR_ref);
11
12 % Measured Noise Generator Values (dB)
13 Pn_dB = [-22 -24 -26 -28 -30 -32 -34 -35 -36];
14
15 % Measured Probability of Bit Error (Pb)
16 Pb = [2457e-4 1960e-4 1424e-4 841e-4 447e-4 ...
17       207e-4 412e-5 155e-5 36e-5];
18
19 % Calculate SNR points corresponding to noise reduction steps
20 SNR_dB = SNR_ref_dB - (Pn_dB - Pn_dB(1));
21
22 % Generate Theoretical Curves
23 SNR_theory_dB = -5:0.05:16;
24 SNR_theory = 10.^(SNR_theory_dB / 10);
25
26 % Note: Because SNR is measured physically via RMS voltage (Ps/Pn)
27 % rather than theoretical Eb/N0, the theoretical Pb equations adjust to:
28 Pb_OOK = qfunc(sqrt(SNR_theory / 2));
29 Pb_BPSK = qfunc(sqrt(SNR_theory));
30
31 % Plot Results
32 figure;
33 semilogy(SNR_theory_dB, Pb_OOK, 'b-', 'LineWidth', 1.5);
34 hold on;
35 semilogy(SNR_theory_dB, Pb_BPSK, 'r--', 'LineWidth', 1.5);
36 semilogy(SNR_dB, Pb, 'ko-', 'LineWidth', 1.3, 'MarkerFaceColor', 'k');
37
38 grid on;
39 grid minor;
40 xlabel('SNR / dB');
41 ylabel('Bit error probability P_b');
42 legend('OOK theory', 'BPSK theory', 'OOK measured', ...
43       'Location', 'southwest');
44 ylim([1e-5 1]);

```